

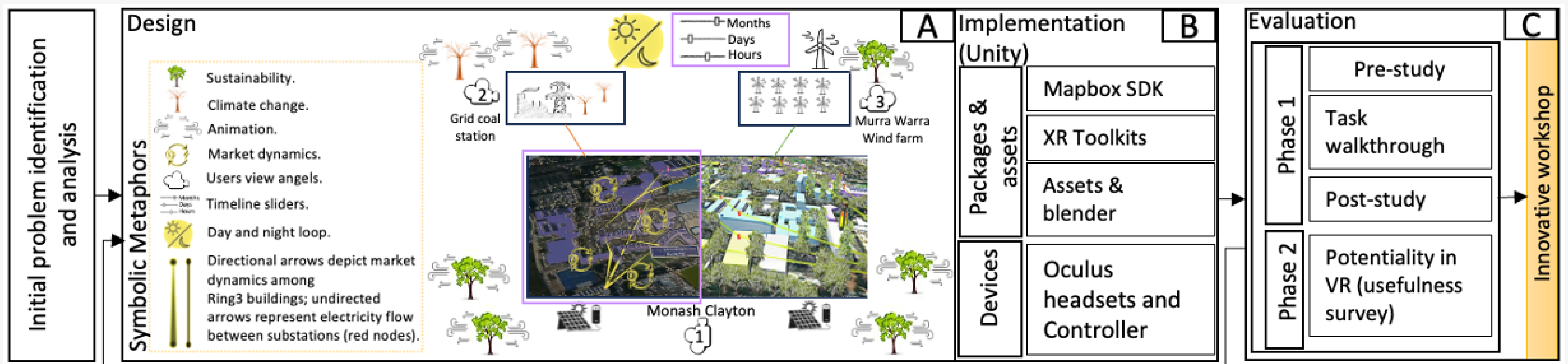


Figure 1: Design science methodology: (A) Metaphorical abstraction design of the Monash Clayton Network, featuring interconnected campus layers, Murra Warra Wind Farm, and aging grid infrastructure where components with purple borders (market dynamics and timelines sliders) are left for future development. (B) Implementation hardware and software. (C) Future evaluation

Abstract

Immersive experiences can increase engagement and improve data understanding. We explore the use of virtual reality to visualise the complexities of electric networks and facilitate a greater understanding of net zero initiatives. We propose that an abstract metaphorical immersive experience can provide an intuitive overview of the complex components of energy networks and how they behave over time in response to fluctuating supply and demand in order to achieve stability and sustainability. This work aims to enable a variety of stakeholders to better understand and associate the complexities in the systems and be aware of the connections over time and space. Using the Monash University electric network system as a case study, we explore the network dynamics and fluctuations over time. The virtual reality experience allows users to view campus electricity assets and local and remote energy sources, as well as witness the dynamics of the network. Thus, the complexities of the energy system and its impact on sustainability can be more readily understood.

Methodology

- The project is framed in accordance with a design science methodology (see Figure 1_A,B,C).
- With graphic abstraction as a pivotal visual design tool known to augment recognition and perception via visual metaphors [1], this study builds upon prior user-centred research [2, 3, 4, 5].

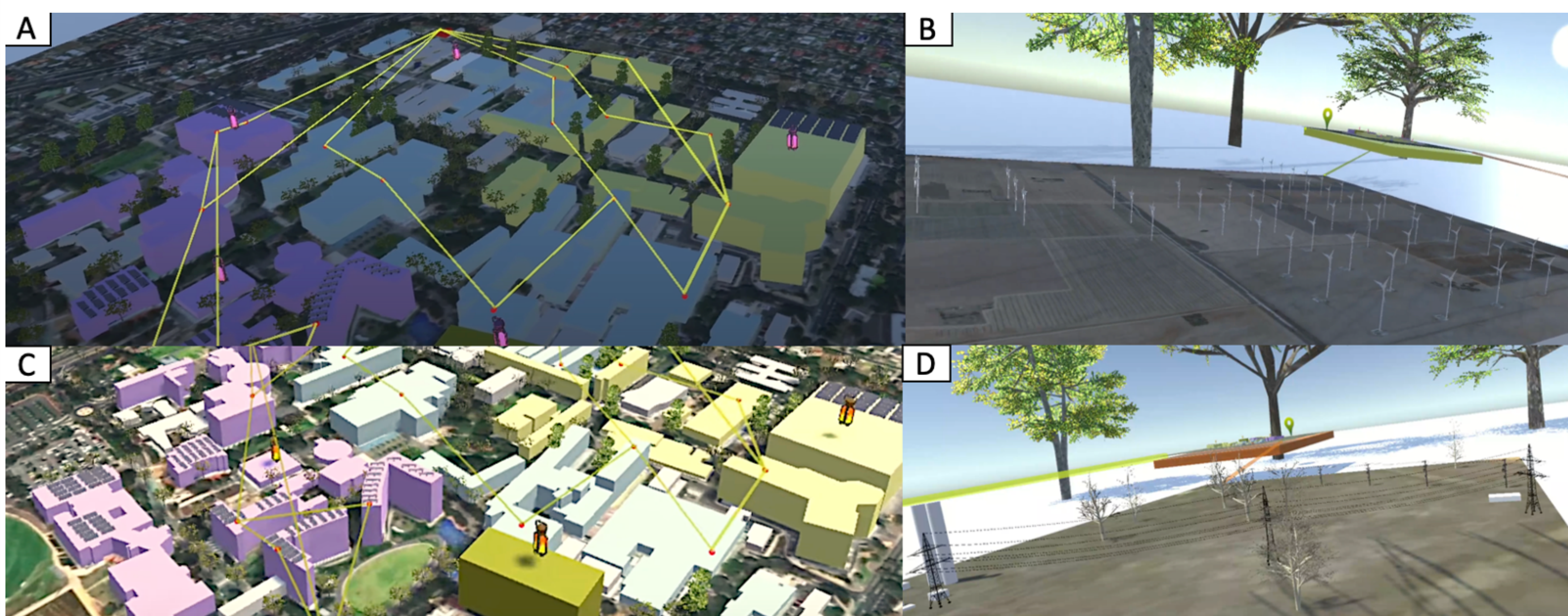


Figure 2: Immersive VR environment illustrating Clayton campus electric network dynamics: (A) Batteries provide stored energy at night. In the early morning, when wind turbines slow down, the connection between Murra Warra and campus becomes thinner. (C) Daytime consumption is powered by solar and wind. (D) During certain periods, the campus relies on grid generation. Off-grid energy means a thin edge.

Contributions

- A metaphorical abstracted visualisation prototype that facilitates stakeholders' understanding of electric network system dynamics (see Figure 2_A,B,C,D).
- An innovative solution that simplifies network complexity, while also taking into account its geographical context.
- Continuing our work in this area will help facilitate the transition to a net zero emissions world, which contributes to the Sustainable Development Goals of the United Nations.

Design and implementation

- Abstracted symbolic metaphors for simplified network complexity.
- The design reflects the fluctuating electric connections over time and space.
- Three visually interactable 3D mini-maps (see Figure 2_A,B,C,D) representing different nodes in the electric network were developed (see Figure 3)
- Detailed immersion in 3D virtual environment.
- Dynamic interactions among three nodes
- Chronological scenarios
 - Morning emphasis on wind turbines
 - Mid-morning sees the traditional grid ensuring reliability amidst heightened demand.
 - Daytime solar panels and wind turbines shine, supplying significant sustainable energy.
 - Nighttime energy reduced batteries provide stored energy to the campus.
- Market dynamics left out for future implementation (see Figure 4).
- As demonstrated in Figure 1_B, the prototype was implemented in Unity where the Mapbox Unity SDK was used to create maps.

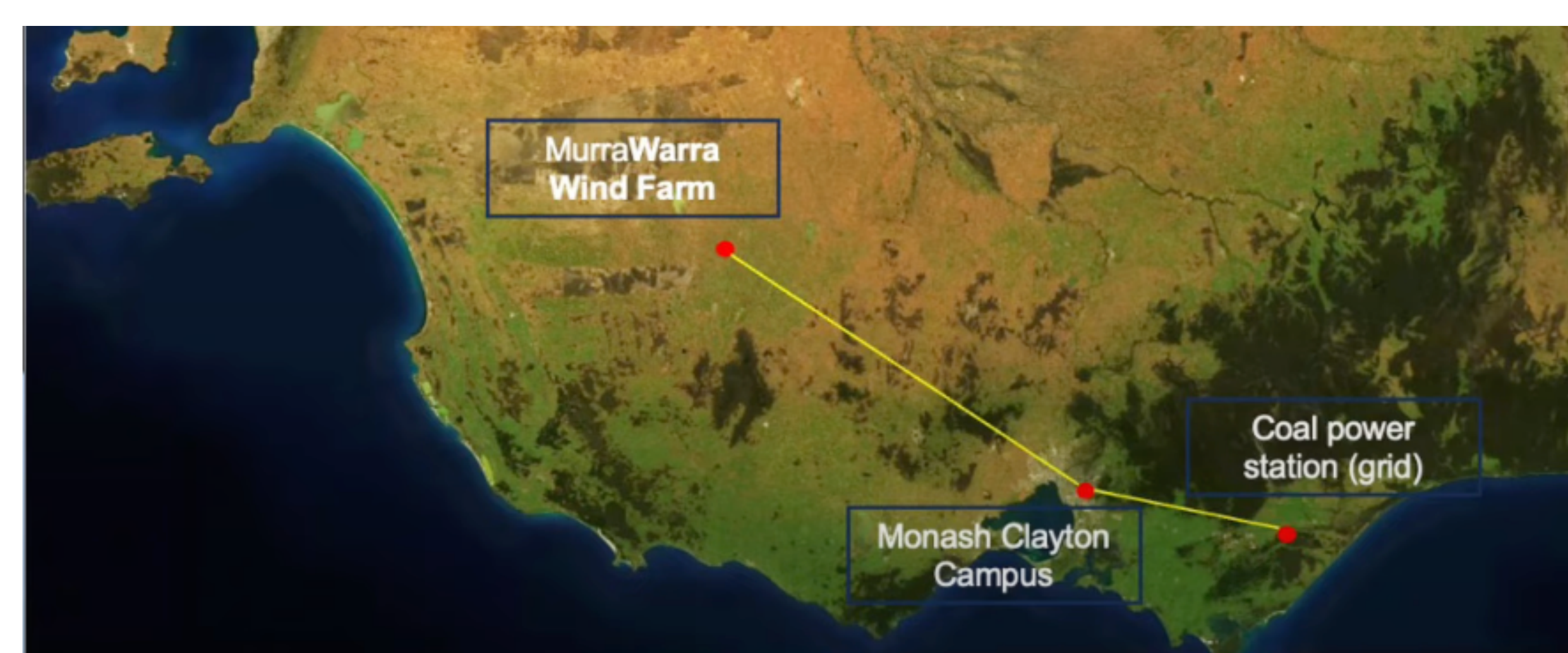


Figure 3: Monash electric network



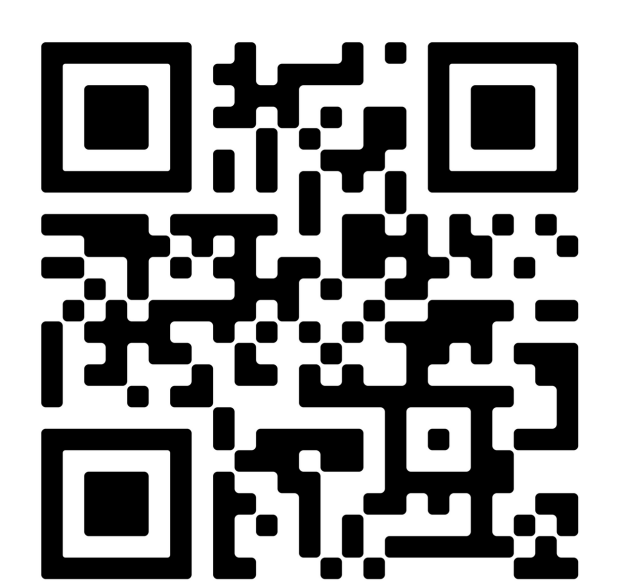
Figure 4: Market dynamics within campus

Conclusion and future directions

Immersive technology provides:

- A unique opportunity for exploring and improving electricity distribution network visualisations.
- It can contribute to better understanding and association with complex systems.

We should explore ways to leverage this technology in the near future to engage a larger range of stakeholders with net zero initiatives.



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